

Power BI – Data Transformation & Data Modeling

E-Commerce Sales Analysis

This assignment will help you explore e-commerce sales data analysis using Power BI.

In this exercise, i have leverage Power BI's capabilities to import, transform, model, and

analyze the provided data.

Answer snip

Data Transformation:

- **Restrict the "List of Orders" table to only the first 500 rows.**
- **Ensure the "Order Date" column in the "List of Orders" table is set to data type 'Date'.**
- **Change the data type of "Amount" and "Target" columns to 'Fixed Decimal Number'.**
- **Format the "CustomerName" column into proper case, ensuring consistent capitalization for each word.**
- **Merge the "State" and "City" columns to create a new column named "Location" in the format 'City, State'.**
- **Create a new custom column named "Profit Margin" as the percentage of "Profit" divided by "Amount".**
- **Add a new conditional column named "Profit Status" based on the values in the**

"Profit" column. The conditions are as follows: if the profit is less than 0, the label

should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the

profit is greater than 0, the label should be "Profit".

Merging Data (Joins):

- **Merge the "List of Orders" and "Order Details" tables into a new single table named**

"Orders Data" based on the "Order ID" relationship.

Handling Missing Data & Duplicate Data:

- **Identify missing values in the data and determine a strategy to address them.**
- **Check for duplicate rows and define a strategy to handle duplicates.**

Sorting and Filtering Data:

- **In the 'Orders Data' table, utilize sorting and filtering techniques on columns like**

Order Date, State or Category to analyze data based on specific criteria:

- ◆ **Sort the orders by Order Date in descending order to analyze recent trends.**

- ◆ **Filter the orders to focus only on a specific state (e.g., Tamil Nadu) for regional analysis.**

Grouping and Aggregating Data:

- **Duplicate the "Order Details" table and calculate the count of each Order ID, average**

profit by Category or total amount by Sub-Category.

- **Duplicate the "Sales Target" table and aggregate the total target amount by Month of**

Order Date.

Data Modeling:

- Establish a relationship between the “List of Orders” and “Order Details” tables using

the ‘Order ID’ column.

- Build a relationship between the “Order Details” and “Sales Target” tables based on

the ‘Category’ column. Click "Manage relationships" and ensure this relationship is

active.

Table.RenameColumns("#Merged city & state",{{"Merged", "Locations"}})				
Order ID	Order Date	CustomerName	Locations	
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	
1 B-25601	4/1/2018	Bharat	Ahmedabadcomma, spaceGujarat	
2 B-25602	4/1/2018	Pearl	Punecomma, spaceMaharashtra	
3 B-25603	4/3/2018	Jahan	Bhopalcomma, spaceMadhya Pradesh	
4 B-25604	4/3/2018	Divsha	Jaipurcomma, spaceRajasthan	
5 B-25605	4/5/2018	Kasheen	Kolkatacomma, spaceWest Bengal	
6 B-25606	4/6/2018	Hazel	Bangalorecomma, spaceKarnataka	
7 B-25607	4/6/2018	Sonakshi	Kashmircomma, spaceJammu and Kashmir	
8 B-25608	4/8/2018	Aarushi	Chennaicomma, spaceTamil Nadu	
9 B-25609	4/9/2018	Jitesh	Lucknowcomma, spaceUttar Pradesh	
10 B-25610	4/9/2018	Yogesh	Patnacomma, spaceBihar	
11 B-25611	4/11/2018	Anita	Thiruvananthapuramcomma, spaceKerala	
12 B-25612	4/12/2018	Shrichand	Chandigarhcomma, spacePunjab	
13 B-25613	4/12/2018	Mukesh	Chandigarhcomma, spaceHaryana	
14 B-25614	4/13/2018	Vandana	Simlacomma, spaceHimachal Pradesh	
15 B-25615	4/15/2018	Bhavna	Gangtokcomma, spaceSikkim	
16 B-25616	4/15/2018	Kanak	Goacomma, spaceGoa	
17 B-25617	4/17/2018	Sagar	Kohimacomma, spaceNagaland	
18 B-25618	4/18/2018	Manju	Hyderabadcomma, spaceAndhra Pradesh	
19 B-25619	4/18/2018	Ramesh	Ahmedabadcomma, spaceGujarat	

Table.AddColumn("#Changed Type1", "Profit Status", each if [Profit] < 0 then "" LOSS "" else if

Order ID	Amount	Profit	Quantity	Category
1 B-25601	1,275.00	-1148	7	Furniture
2 B-25601	66.00	-12	5	Clothing
3 B-25601	8.00	-2	3	Clothing
4 B-25601	80.00	-56	4	Electronics
5 B-25602	168.00	-111	2	Electronics
6 B-25602	424.00	-272	5	Electronics
7 B-25602	2,617.00	1151	4	Electronics
8 B-25602	561.00	212	3	Clothing
9 B-25602	119.00	-5	8	Clothing
10 B-25603	1,355.00	-60	5	Clothing
11 B-25603	24.00	-30	1	Furniture
12 B-25603	193.00	-166	3	Clothing
13 B-25603	180.00	5	3	Clothing
14 B-25603	116.00	16	4	Clothing
15 B-25603	107.00	36	6	Clothing
16 B-25603	12.00	1	2	Clothing
17 B-25603	38.00	18	1	Clothing
18 B-25604	65.00	17	2	Clothing

Table.TransformColumnTypes("#changed first row as header",{{"Month of Order Date", type date},

Month of Order Date	Category	Target
4/18/2025	Furniture	10,400.00
5/18/2025	Furniture	10,500.00
6/18/2025	Furniture	10,600.00
7/18/2025	Furniture	10,800.00
8/18/2025	Furniture	10,900.00
9/18/2025	Furniture	11,000.00
10/18/2025	Furniture	11,100.00
11/18/2025	Furniture	11,300.00
12/18/2025	Furniture	11,400.00
1/19/2025	Furniture	11,500.00
2/19/2025	Furniture	11,600.00
3/19/2025	Furniture	11,800.00
4/18/2025	Clothing	12,000.00
5/18/2025	Clothing	12,000.00
6/18/2025	Clothing	12,000.00
7/18/2025	Clothing	14,000.00
8/18/2025	Clothing	14,000.00
9/18/2025	Clothing	14,000.00
10/18/2025	Clothing	16,000.00

changed target type to Fixed Decimal N

Table.SelectRows("#descending order", each ([Locations] = "Chennaicomma, spaceTamil Nadu"))

Order ID	Order Date	CustomerName	Locations	Order Details
1 B-26081	3/22/2019	Aarushi	Chennaicomma, spaceTamil Nadu	Table
2 B-26018	2/14/2019	Aarushi	Chennaicomma, spaceTamil Nadu	Table
3 B-26008	2/9/2019	Kalyani	Chennaicomma, spaceTamil Nadu	Table
4 B-25860	11/15/2018	Akshay	Chennaicomma, spaceTamil Nadu	Table
5 B-25788	9/21/2018	Dinesh	Chennaicomma, spaceTamil Nadu	Table
6 B-25716	7/11/2018	Surabhi	Chennaicomma, spaceTamil Nadu	Table
7 B-25698	6/23/2018	Amisha	Chennaicomma, spaceTamil Nadu	Table
8 B-25608	4/8/2018	Aarushi	Chennaicomma, spaceTamil Nadu	Table

Table.Group("#orders count", {"Order ID"}, {"Orders count", each Table.RowCount(_), Int64.Type})

	Order ID	Orders count
1	B-25601	4
2	B-25602	5
3	B-25603	8
4	B-25604	2
5	B-25605	1
6	B-25606	1
7	B-25607	1
8	B-25608	4
9	B-25609	2
10	B-25610	6
11	B-25611	1
12	B-25612	1
13	B-25613	1
14	B-25614	2
15	B-25615	1
16	B-25616	4
17	B-25617	1
18	B-25618	2
19	B-25619	1

Table.RenameColumns("#grouped by category & sum of amount", {"Count", "sumof amount"})

	Sub-Category	sumof amount
1	Bookcases	56861
2	Stole	18546
3	Hankkerchief	14608
4	Electronic Games	39168
5	Phones	46119
6	Saree	53511
7	Trousers	30039
8	Chairs	34222
9	Kurti	3361
10	T-shirt	7382
11	Shirt	7555
12	Leggings	2106
13	Tables	22614
14	Printers	58252
15	Accessories	21728
16	Furnishings	13484
17	Skirt	1946

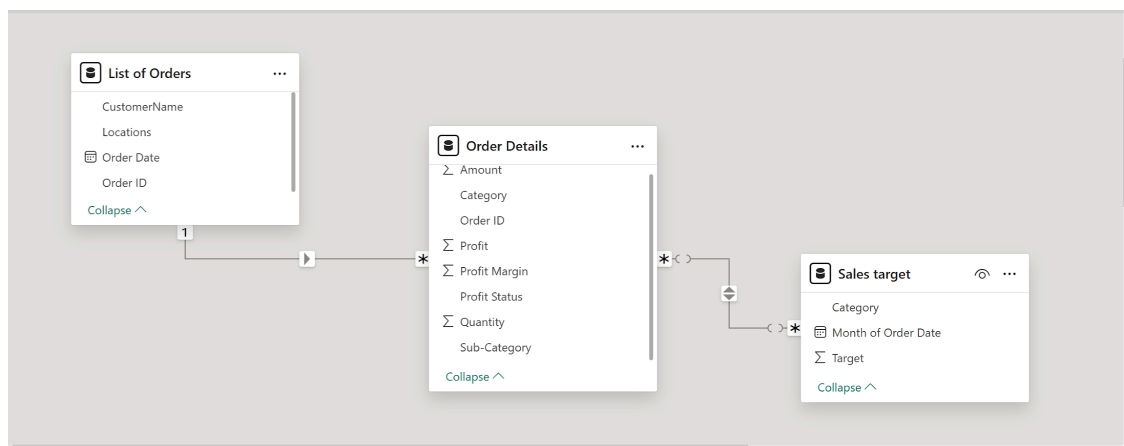
Furniture	9.456790123	
Clothing	11.76290832	
Electronics	34.07142857	

ased on top 1000 rows

Table: RenameColumns("#grouped by total target by month of order date",{"Count", "Total target by month of order date"})

	Month of Order Date	1.2 Total target by month
	Valid 100%	Valid 100%
	Error 0%	Error 0%
	Empty 0%	Empty 0%
1	4/18/2025	31400
2	5/18/2025	31500
3	6/18/2025	31600
4	7/18/2025	33800
5	8/18/2025	33900
6	9/18/2025	34000
7	10/18/2025	36100
8	11/18/2025	36300
9	12/18/2025	36400
10	1/19/2025	43500
11	2/19/2025	43600
12	3/19/2025	43800

profiling based on top 1000 rows



Thank you

Valli.